Topology-Invariant Error Dynamics and Detection-Free Cable-Fault Tolerance

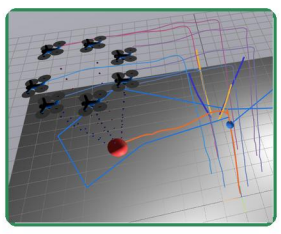
in Decentralized Multi-Quadrotor Cooperative Transport

When cables snap mid-flight, the remaining drones absorb the fault and keep tracking the payload — with no fault detection, no communication, and no reconfiguration.

No fault detection | No communication | No reconfiguration | 43.6 cm RMSE

1 Three-stage recovery sequence

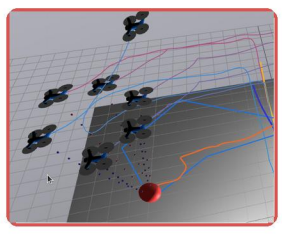
01



01 Nominal operation

Seven quadrotors carry a cable-suspended payload in heptagonal formation. No drone knows the team size or other agents' states.

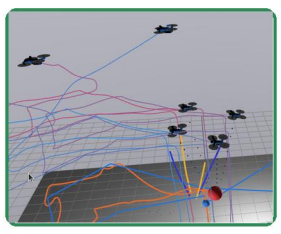
02



02 Cable snap

At t = 7 s, cable 1 snaps. The freed drone accelerates away. Remaining drones get a sudden 17% load-share increase.

03



03 Stable recovery

After 3 cascading snaps (t = 7, 12, 14 s), 4 surviving drones keep tracking with bounded error (43.6 cm RMSE).

2 Why the system stays stable

Topology-invariant error dynamics

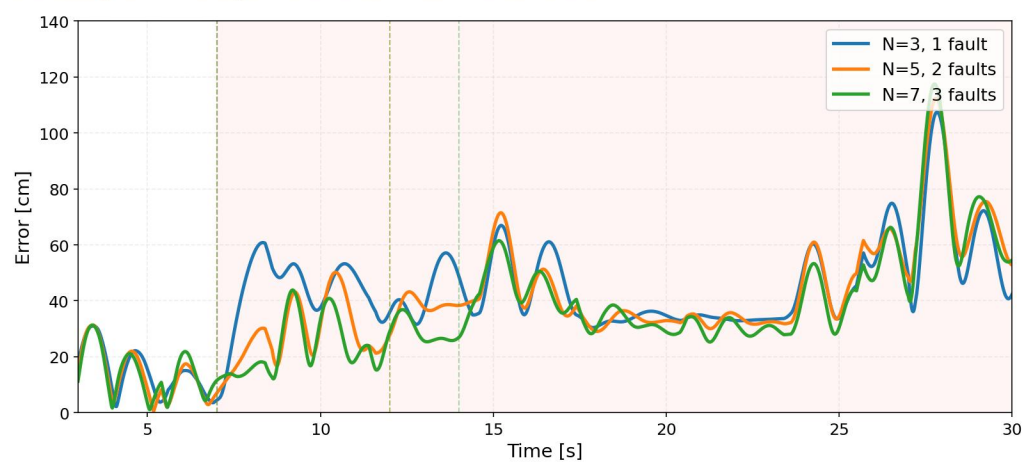
Cable snaps change the disturbance input d_i(t), but not the closed-loop matrices (A, B). The error dynamics remain stable across all cable topologies.

$$z_dot = A\ z + B\ d_i(t)$$

- **Proposition 1**
  A, B depend only on PID gains (kp, kd, ki) — independent of N, payload mass, or cable topology.
- **Common Lyapunov**
  A single Lyapunov function works across all cable topologies — no switching needed.
- **Design tools**
  Cost ratio: ρ_FT = N/(N−k).
  Thrust margin: f_max > (Nm_Q+m_L)g/(N−k) + f_track.
- **Oracle comparison**
  A centralized controller with perfect fault knowledge improves RMSE by only 0.7%.

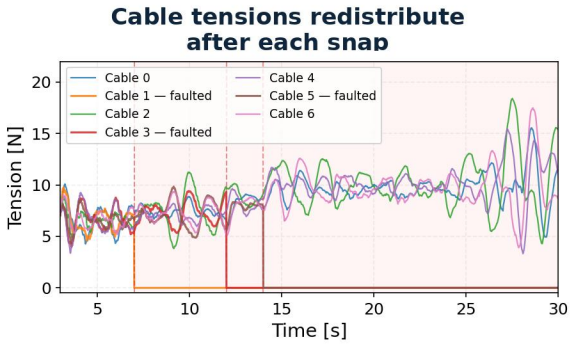
<3% MC variation | 0.7% oracle gain

Tracking error stays bounded after each fault

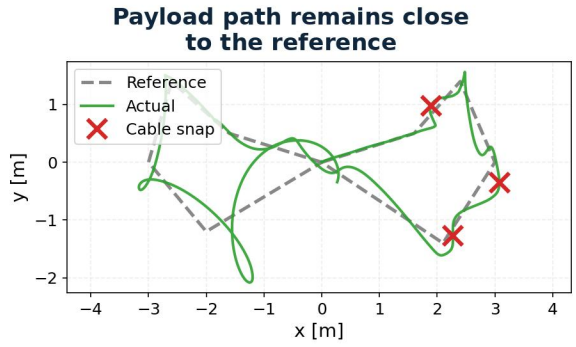


3 Evidence from the multibody simulations

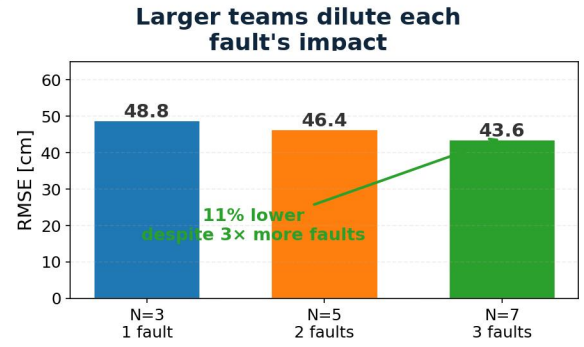
Cable tensions redistribute after each snap



Payload path remains close to the reference

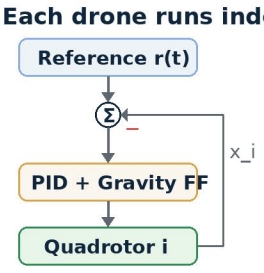


Larger teams dilute each fault's impact



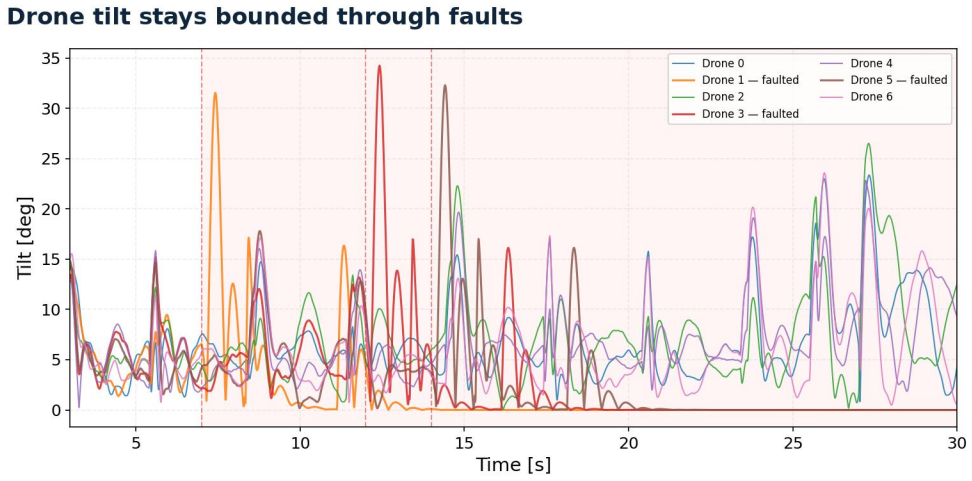
4 Decentralized control architecture

Each drone runs independently



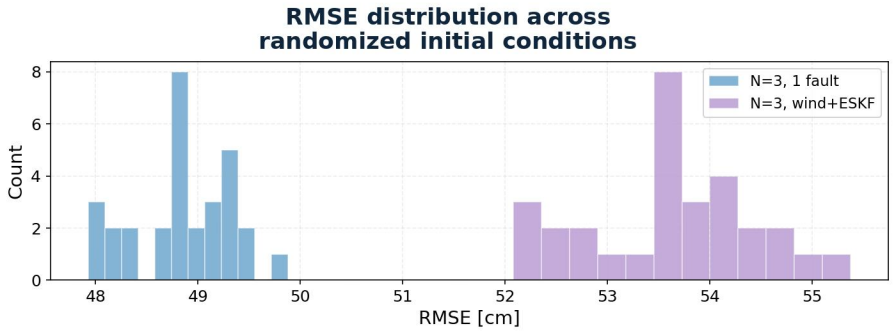
- **N-agnostic**
  Controller does not know the team size N
- **No communication**
  No inter-drone data exchange required
- **Gravity feedforward**
  Compensates payload weight share: m_L g / N
- **Cable-mediated only**
  Coupling through physics, not information

Drone tilt stays bounded through faults

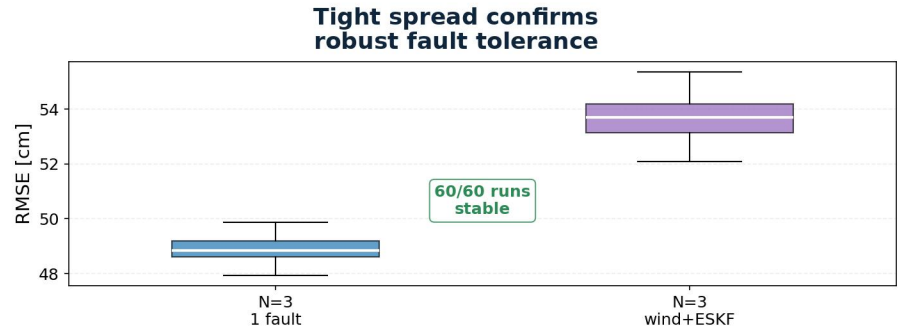


5 Monte Carlo robustness (90 runs)

RMSE distribution across randomized initial conditions



Tight spread confirms robust fault tolerance



Key takeaway: a local N-agnostic PID + gravity-feedforward controller can tolerate cascading cable snaps without communication or explicit fault detection.